

Akshay Shankar

COMPUTATIONAL PHYSICIST

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Research Focus: Understanding the behaviour of ultra-cold matter through numerical simulations to enable practical quantum technologies.

Summary

PhD Student at Ghent University specializing in Tensor Network simulations for one-dimensional quantum gases in the continuum. Experienced in developing open-source software for high-performance numerical algorithms (Julia/Python) to study quantum many-body systems. Also experienced in building interactive visualizations and user interfaces using front-end web development tools and game engines.

Skills

Scientific Programming	Julia (TensorKit.jl, MPSKit.jl), Python (NumPy, SciPy), C++, Git, Linux
Numerical Methods	Tensor Networks (DMRG, VUMPS, TDVP), Quantum Monte Carlo (SSE), Mean-field theory (Gutzwiller, Cluster-MFT)
Visualization & UI	Godot Engine (GDScript), Blender, Inkscape, Front-end (HTML, CSS, JavaScript)

Research Experience

Ph.D. Project: Finite-element matrix product states for continuum models

Ghent University

SUPERVISED BY PROF. KAREL VAN ACOLEYEN AND PROF. JUTHO HAEGEMAN | JULIA

Sept 2023 - Present

- Developed a theoretical framework to utilize Matrix Product States (MPS) with *non-orthogonal* single-particle orbitals to study quantum gases in one dimension.
- Applied the framework to capture ground state physics of the Lieb-Liniger model using a finite-element basis and a multi-grid optimization scheme to efficiently reach the continuum limit.
- Developed a performant, open-source numerical implementation of the finite-element Matrix Product States (FE-MPS) formalism involving custom DMRG code built on top of MPSKit.jl.

M.S. Thesis: Low-temperature phases of interacting bosons in a lattice

IISER Mohali & University of Stuttgart

SUPERVISED BY DR. SANJEEV KUMAR AND PROF. TILMAN PFAU | JULIA, PYTHON

June 2022 - Apr. 2023

- Implemented various many-body algorithms (Cluster Mean-Field Theory, Stochastic Series Expansion, Neural Quantum States) to map the phase diagram of the extended Bose-Hubbard model on a square lattice.
- Collaborated with experimentalists at the University of Stuttgart to guide the exploration of supersolid phases arising from Dysprosium atoms with dipole-dipole interactions in an optical lattice.

Research Internship: Simulating BEC dynamics in one dimension

IIT Kharagpur (Remote)

SUPERVISED BY DR. VISHWANATH SHUKLA | JULIA, FORTRAN90

May 2021 - Aug. 2021

- Developed a 1D Gross-Pitaevskii Equation (GPE) solver to simulate ground-state physics and real-time dynamics of Bose-Einstein Condensates.
- Investigated parallel computing frameworks (MPI, OpenMP) to optimize performance.

Collaborative projects

Game Jam Competitions

TEAM LEAD & LEAD PROGRAMMER | GODOT, GIT

2023 - Present

- Led teams of up to 5 to conceptualize and develop interactive game prototypes within strict 48-96 hour deadlines.
- Utilized Git for version control, ensuring project stability under rapid development cycles.
- Managed the programming life-cycle from designing the overall code architecture to final deployment on itch.io through Github Actions.

ThePhysicsHub

CO-FOUNDER | JAVASCRIPT, P5.JS

2020-2021

- Founded a global community of students to maintain a repository of interactive simulations for educational purposes.
- Established a standardized user interface and developed several simulations demonstrating concepts of classical mechanics.

Publications & Preprints

Finite-Element Matrix Product States for Continuum Models in One Dimension

AKSHAY SHANKAR, KAREL VAN ACOLEYEN, AND JUTHO HAEGEMAN
<https://arxiv.org/abs/2606.14873>

Arxiv
Jun. 2026

Open Source Software

TentMPS.jl

IMPLEMENTATION OF THE FE-MPS FORMALISM DEVELOPED IN ARXIV:2606.14783.

2026-Present

LiebLinigerBetheAnsatz.jl

IMPLEMENTATION OF THE BETHE ANSATZ SOLUTION FOR THE LIEB-LINIGER MODEL.

2025-Present

Education

Ghent University

PH.D. IN PHYSICS

- Supervised by Prof. Karel Van Acoleyen and Prof. Jutho Haegeman.

Gent, Belgium
Oct. 2023 - Present

Indian Institute of Science Education and Research (IISER), Mohali

B.S.-M.S. INTEGRATED DEGREE (MAJOR IN PHYSICS)

- Cumulative Performance Index (CPI): 10.0/10.0

Punjab, India
Aug. 2018 - June. 2023

Honors & Awards

2023 **President's Gold Medal**, for the best academic performance in the class of 2023.

IISER Mohali

2019 **CNR Rao Foundation Award**, for obtaining 10.0 SPI in the 2nd semester.

IISER Mohali

2018 **Innovation in Science Pursuit for Inspired Research - Scholarship for Higher Education (INSPIRE-SHE)**,
for top 1% performance in AISSCE.

DST India

Conferences & Workshops

2026 **Challenges of Low-Dimensional Quantum Gases: Brazilian-German Wilhelm and Else Heraeus Seminar**, presented a poster titled "Finite element matrix product states for continuous systems"

ICTP-SAIFR, Sao
Paulo, Brazil

2024 **Les Houches Predoc school on cold atoms**, presented a poster titled "RadialMPS: A novel scheme to study 2D Bose gases in the continuum"

Les Houches,
France

2024 **European tensor network school**, presented a poster titled "RadialMPS: A novel scheme to study 2D Bose gases in the continuum"

Padua, Italy